

CUSTOMER DATA ANALYSIS

USING SQL & POWER BI

Business Intelligence & Data Analytics Report

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Prepared for Business Intelligence Analysis and Strategic Decision-Making

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3. Executive Summary

This report presents a comprehensive business intelligence analysis of customer shopping behavior conducted on a dataset of **1,000** transaction records collected across ten shopping malls. The analysis was performed in two stages: **(1)** data preparation and exploratory analysis using MySQL and SQL, followed by **(2)** the development of two interactive Power BI dashboards—an Executive Overview dashboard and a Business Insights dashboard—using Power Query for data transformation and DAX for KPI and measure calculations.

The engagement was undertaken to help business stakeholders understand who their customers are, what they purchase, how they pay, and where they shop, so that marketing, merchandising, and mall-partnership decisions can be made on evidence rather than assumption. Across the dataset, total revenue reached approximately **2.37M** currency units, generated from roughly **3,000** units sold across eight product categories. The average order value and average customer spend were each approximately **2,37K** with an average of **2.98** items purchased per transaction at an average unit price of approximately **653.02**.

Key findings show that customers in the **50+ age group** are the single largest contributor to both transaction volume and revenue, that female customers account for the majority of transactions and spending, that **Clothing** is the leading product category by both quantity and revenue, that cash remains the most frequently used payment method, and that the **Mall of Istanbul** is the top-performing shopping location by quantity sold, customer footfall, and revenue. The following sections present detailed SQL analysis, interactive dashboard insights, business interpretations, and strategic recommendations to support data-driven decision-making.

4. Project Overview

The Customer Data Analysis project was designed to convert raw, transaction-level retail data into structured, decision-ready business intelligence. The source data captures individual customer purchases made across multiple shopping malls, including details such as the customer's gender and age, the product category purchased, the quantity and price of items bought, the payment method used, the date of the transaction, and the shopping mall where the purchase took place.

The project followed a standard business intelligence workflow. The raw data was first loaded into a MySQL relational database and organized into a single structured customer table. SQL was then used to validate data quality, identify missing values and duplicate records, standardize text fields, and create two analytical fields: Revenue and Age Group.

Once the data was verified as clean and consistent, eleven core business questions were answered directly in SQL to establish a factual baseline of customer and sales patterns. This SQL-based analysis was then extended into Power BI, where Power Query was used to import and transform the data, and DAX was used to build KPI measures. The results were assembled into two dashboards: an Executive Overview dashboard aimed at senior stakeholders who need a high-level summary of performance, and a Business Insights dashboard aimed at analysts and category managers who need to explore customer segments and shopping mall performance in more depth.

5. Business Objectives

The analysis was guided by a defined set of business objectives, each intended to answer a specific question that the business had about its customers and its sales performance:

- Understand how shopping activity, product quantity, and revenue are distributed across gender to support gender-focused marketing and merchandising decisions.
- Understand how shopping activity, product quantity, and revenue vary by customer age group to identify the most valuable age segments.
- Identify which product categories are purchased most frequently by each gender and age group, to guide product placement and promotional targeting.
- Evaluate customer payment preferences and their relationship with customer demographics, to inform payment infrastructure and checkout strategy.
- Assess the performance of individual shopping malls in terms of quantity sold, customer footfall, and revenue generated, to support mall-partnership and resource-allocation decisions.
- Establish a set of core KPIs - average order value, average customer spend, average quantity per transaction, and average unit price - that can be tracked on an ongoing basis to monitor business health.
- Deliver the findings through interactive, self-service Power BI dashboards that allow non-technical stakeholders to explore the data by gender, age group, product category, shopping mall, and payment method.

6. Dataset Description

6.1 Dataset Overview

The dataset used for this analysis is a customer shopping dataset stored in a MySQL database named **customer_data_analysis**, containing a single table named **customer**. Each row in the table represents one line-item transaction, capturing who the customer was, what they bought, how much they bought, what they paid, how they paid, when the purchase occurred, and at which shopping mall the purchase took place. The data was imported into MySQL Workbench from a CSV source file using the Table Data Import Wizard, after which it was cleaned, validated, and enriched using SQL before being loaded to Power BI for dashboard development.

6.2 Number of Records

The customer table contains **1,000** individual transaction records, with no records removed during data validation, as the duplicate and null-value checks performed during data preparation returned no data quality issues requiring row removal.

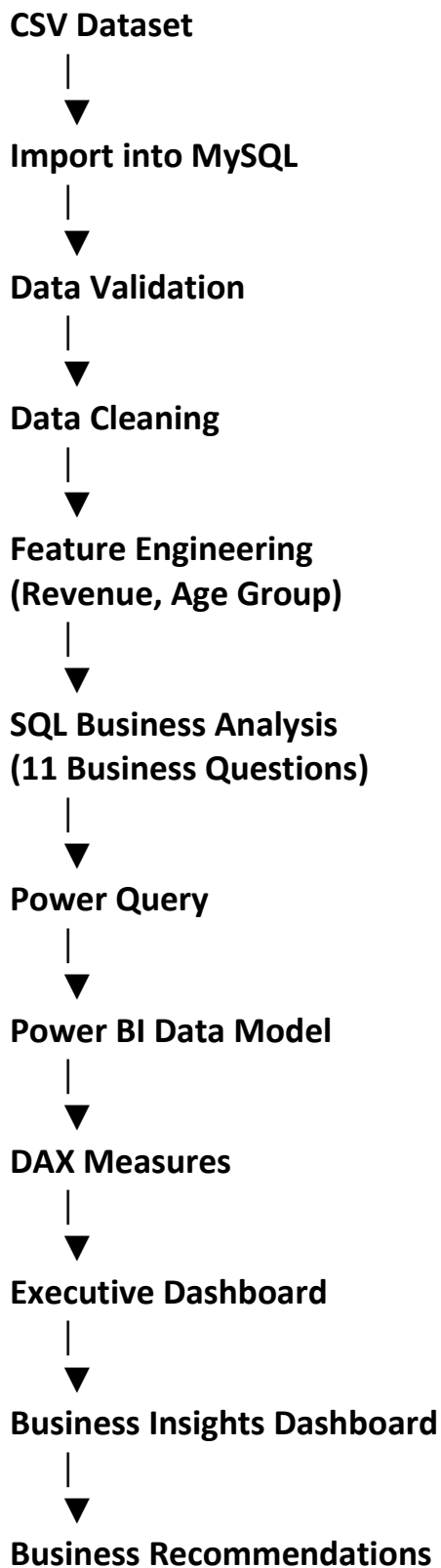
6.3 Number of Columns

The raw dataset contains ten columns as imported from the source file. During data preparation, two additional analytical columns - **revenue** and **age_group** - were engineered using SQL, bringing the total number of columns available for analysis to twelve.

6.4 Column Descriptions

Column Name	Data Type	Description
invoice_no	VARCHAR(20)	Unique identifier assigned to each customer invoice or transaction.
customer_id	VARCHAR(20)	Unique identifier assigned to each customer.
gender	VARCHAR(20)	Gender of the customer (Female / Male).
age	INT	Age of the customer in years at the time of purchase.
category	VARCHAR(50)	Product category purchased, e.g. Clothing, Shoes, Technology, Cosmetics.
quantity	INT	Number of units purchased in the transaction.
price	DECIMAL(10,2)	Unit price of the product purchased, in currency units.
payment_method	VARCHAR(30)	Method used to pay for the transaction (Cash, Credit Card, Debit Card).
invoice_date	VARCHAR(20)	Date on which the transaction was recorded.
shopping_mall	VARCHAR(100)	Shopping mall location at which the transaction took place.
revenue (derived)	DECIMAL(12,2)	Calculated field representing the total transaction value (Quantity × Price).
age_group (derived)	VARCHAR(20)	Calculated field categorizing customers into age groups (Teen, 20–30, 31–40, 41–50, and 50+).

7. Data Analysis Workflow



8. Data Preparation

Before any analysis was performed, the dataset was subjected to a structured data preparation process to ensure that all downstream business conclusions were based on accurate, consistent, and complete data. This process was carried out primarily in SQL, with additional data transformation performed in Power Query at the point of loading the data into Power BI.

8.1 Data Validation and Quality Checks

The total record count of the customer table was first confirmed using a row-count query to establish a baseline of 1,000 records. Each of the ten source columns was then checked for null or missing values using a single validation query that tested every column simultaneously. This check confirmed that the dataset contained no missing values in any field, meaning that every transaction record was complete across invoice number, customer ID, gender, age, category, quantity, price, payment method, invoice date, and shopping mall.

8.2 Duplicate Record Checks

Duplicate transactions were checked by grouping records on the invoice number field and counting occurrences greater than one. This check confirmed that every invoice number in the dataset is unique, indicating that there is no duplication of transaction records that would otherwise inflate quantity or revenue totals.

8.3 Categorical Value and Range Verification

To confirm that categorical fields did not contain inconsistent or unexpected entries, distinct-value checks were run against gender, product category, payment method, and shopping mall. This step is a standard control that verifies, for example, that gender contains only valid values and that no category or mall name is duplicated under a different spelling. In parallel, the numeric fields - age, quantity, and price - were checked for their minimum and maximum values to confirm that the data fell within a realistic and expected range, with no negative or implausible values present.

8.4 Data Type Verification and Text Standardization

Each column's data type was verified against its intended use during table creation - identifiers and categorical fields as VARCHAR, numeric quantities as INT, and monetary values as DECIMAL - ensuring that calculations such as revenue could be performed accurately. To eliminate formatting inconsistencies that can arise from CSV imports, leading and trailing spaces were removed from all key text fields, including gender, category, payment method, and shopping mall, using SQL's TRIM function.

8.5 Data Transformation and Feature Engineering

Two new fields were engineered directly in SQL to support business analysis. A revenue column was added and populated as the product of quantity and price, giving a single measure of transaction value that could be aggregated across gender, age group, category, payment method, and shopping

mall. An `age_group` column was also added and populated using a SQL CASE expression that classifies each customer into one of five bands - Teen, 20-30, 31-40, 41-50, and 50+ - allowing age-based patterns to be analyzed without repeatedly re-bucketing raw age values.

When the cleaned and enriched dataset was brought into Power BI, Power Query was used to confirm column data types, apply consistent naming, and load the table into the Power BI data model, after which DAX was used to define the KPI measures that appear on the two dashboards. Because all cleaning, validation, and transformation had already been completed in SQL, the Power Query step focused on ensuring the model was structured correctly for reporting rather than on further data cleansing.

9. SQL Analysis Methodology

- SQL was used as the primary analytical tool to explore customer purchasing behavior.
 - Data validation queries ensured the dataset contained no missing or duplicate records.
 - Aggregate functions such as **COUNT()**, **SUM()**, **AVG()**, and **ROUND()** were used to calculate business metrics.
 - **GROUP BY** and **ORDER BY** clauses were applied to summarize and rank business performance.
 - A **CASE** expression was used to categorize customers into age groups.
 - Revenue calculations were performed using SQL expressions based on product quantity and price.
 - Each SQL query was designed to answer a specific business question.
 - The results were interpreted from a business perspective to support data-driven decision-making.
-

10. SQL Business Analysis

Eleven core business questions were answered using SQL against the cleaned customer dataset. Each query analyzes transaction-level data to uncover meaningful patterns in customer behavior, product performance, payment preferences, and shopping mall performance. For each business question, the SQL query, SQL output, and business interpretation are presented in the same sequence in which they were executed during the project.

Business Question 1: How is the shopping distribution according to gender?

SQL Query:

```
SELECT gender,  
       COUNT(invoice_no) AS total_transactions  
FROM customer  
GROUP BY gender;
```

SQL Output:

Gender	Total Transactions
Male	380
Female	620

The SQL query groups customer transactions by gender and returns the total number of transactions recorded for each gender.

Business Interpretation:

Female customers account for 620 transactions (62%), compared with 380 transactions (38%) for male customers, making them the dominant customer segment. This suggests that marketing campaigns, loyalty programs, and product assortment strategies should prioritize the preferences of female customers while also developing targeted initiatives to increase engagement among male customers.

Business Question 2: Which gender did we sell more products to?

SQL Query:

```
SELECT gender,  
       SUM(quantity) AS total_products_sold  
FROM customer  
GROUP BY gender;
```

SQL Output:

Gender	Total Products Sold
Male	1128
Female	1853

This query groups customer transactions by gender and calculates the total quantity of products purchased by each group. It helps identify which customer segment contributes more to the overall product sales volume.

Business Interpretation:

The analysis shows that **female customers purchased 1,853 products**, compared to **1,128 products purchased by male customers**. This indicates that female customers contributed a higher share of total product sales volume. Understanding purchasing behavior by gender enables businesses to optimize inventory planning, design targeted marketing campaigns, and tailor product offerings to better meet customer demand.

Business Question 3: Which gender generated more revenue?**SQL Query:**

```
SELECT gender,  
       ROUND(SUM(revenue),2) AS total_revenue  
FROM customer  
GROUP BY gender;
```

SQL Output:

Gender	Total Revenue
Male	955486.66
Female	1408881.11

This query groups customer transactions by gender and calculates the total revenue generated by each group using the revenue column. It helps identify which customer segment contributes more to the company's overall sales revenue.

Business Interpretation:

The analysis shows that **female customers generated a total revenue of 1,408,881.11**, compared to **955,486.66** generated by male customers. This indicates that female customers contributed a significantly higher share of the company's total revenue. These insights help businesses identify their most valuable customer segment, optimize marketing investments, personalize promotional campaigns, and make data-driven decisions to maximize revenue growth.

Business Question 4: What is the distribution of purchase categories relative to gender?

SQL Query:

```
SELECT category,
       gender,
       SUM(quantity) AS total_quantity
FROM customer
GROUP BY category, gender
ORDER BY category;
```

SQL Output:

Category	Gender	Total Quantity
Books	Female	112
Books	Male	84
Clothing	Female	612
Clothing	Male	365
Cosmetics	Female	305
Cosmetics	Male	168

Note: The complete SQL output includes all product categories and is summarized below for readability.

This query groups customer purchases by **product category** and **gender**, then calculates the total quantity sold for each combination. It helps compare purchasing patterns across different product categories and identify which categories are most popular among male and female customers.

Business Interpretation:

The analysis shows that **Clothing** is the highest-selling product category for both genders, with **612 units** purchased by female customers and **365 units** by male customers. Female customers purchased more products than male customers across all displayed categories, indicating stronger purchasing activity. These insights help businesses optimize inventory levels, improve category planning, and design targeted marketing campaigns based on customer preferences.

Business Question 5: How is the shopping distribution according to age?

SQL Query:

```
SELECT age_group,
       COUNT(invoice_no) AS total_transactions
FROM customer
GROUP BY age_group;
```

SQL Output:

Age Group	Total Transactions
20-30	213
31-40	184
41-50	196
50+	367
Teen	40

This query groups customer transactions by **age group** and counts the total number of transactions in each group. It helps identify which age segment contributes the highest shopping activity and supports customer segmentation analysis.

Business Interpretation:

The analysis shows that customers aged **50 and above** recorded the highest number of transactions (**367**), followed by the **20–30** age group (**213**). The **Teen** age group had the fewest transactions (**40**). These insights help businesses understand customer shopping behavior across different age segments, enabling more effective customer segmentation, targeted marketing campaigns, and age-specific promotional strategies.

Business Question 6: Which age category did we sell more products to?

SQL Query:

```
SELECT age_group,  
       SUM(quantity) AS total_products_sold  
FROM customer  
GROUP BY age_group;
```

SQL Output:

Age Group	Total Products Sold
50+	1057
20–30	666
41–50	623
31–40	520
Teen	115

This query groups customer purchases by **age group** and calculates the total quantity of products sold for each segment. It helps identify which age group contributes the highest product sales volume and provides insights into purchasing behavior across different customer age categories.

Business Interpretation:

The analysis shows that customers aged **50 and above** purchased the highest number of products (**1,057 units**), followed by the **20–30** age group (**666 units**), **41–50** (**623 units**), and **31–40** (**520 units**). The **Teen** segment recorded the lowest purchase volume (**115 units**). These findings indicate that older customers are the primary contributors to product sales, helping businesses prioritize inventory management, customer segmentation, and targeted marketing strategies for high-value age groups.

Business Question 7: Which age category generated more revenue?

SQL Query:

```
SELECT age_group,  
       ROUND(SUM(revenue),2) AS total_revenue  
FROM customer  
GROUP BY age_group;
```

SQL Output:

Age Group	Total Revenue
50+	901,446.83
20–30	497,201.70
41–50	485,685.27
31–40	408,962.49
Teen	72,071.48

*This query groups customer transactions by **age group** and calculates the total revenue generated by each segment. It helps identify which age category contributes the highest sales revenue and provides valuable insights into customer purchasing value across different age groups.*

Business Interpretation:

The analysis shows that customers aged **50 and above** generated the highest revenue (**901,446.83**), followed by the **20–30** age group (**497,201.70**) and **41–50** (**485,685.27**). The **Teen** segment generated the lowest revenue (**72,071.48**). These findings indicate that the 50+ customer segment contributes the greatest business value, enabling organizations to prioritize customer retention, targeted marketing campaigns, and personalized promotional strategies for this high-value demographic.

Business Question 8: What is the distribution of purchase categories relative to age group?

SQL Query:

```
SELECT category,
       age_group,
       SUM(revenue) AS total_revenue
FROM customer
GROUP BY category, age_group
ORDER BY category;
```

SQL Output:

Category	Age Group	Total Revenue
Books	20–30	1,954.35
Books	31–40	2,439.15
Books	41–50	2,378.55
Books	50+	3,423.90
Books	Teen	378.75
Clothing	20–30	248,766.32

Note: The complete SQL output includes all category and age group combinations and has been summarized here for readability.

This query groups customer purchases by **product category** and **age group**, then calculates the total revenue generated for each combination. It helps identify which product categories perform best within different age segments and supports age-based customer and product analysis.

Business Interpretation:

The analysis reveals that **Clothing** generated the highest revenue across multiple age groups, particularly among customers aged **50+** and **20–30**. Revenue contributions vary across product categories and age segments, indicating that customer preferences differ by demographic. These insights enable businesses to optimize inventory planning, develop age-specific marketing campaigns, and improve product assortment strategies to maximize revenue.

Business Question 9: Does the payment method have a relation with other columns (gender)?

SQL Query:

```
SELECT payment_method,
       gender,
       COUNT(invoice_no) AS total_transactions
FROM customer
GROUP BY payment_method, gender
ORDER BY payment_method;
```

SQL Output:

Payment Method	Gender	Total Transactions
Cash	Female	268
Cash	Male	152
Credit Card	Female	222
Credit Card	Male	144
Debit Card	Female	130
Debit Card	Male	84

This query groups customer transactions by **payment method** and **gender**, then counts the total number of transactions for each combination. It helps analyze whether payment preferences differ between male and female customers and identifies the most frequently used payment methods.

Business Interpretation:

The analysis shows that **Cash** was the most frequently used payment method for both female (**268 transactions**) and male (**152 transactions**) customers, followed by **Credit Card** and **Debit Card**. Female customers recorded a higher number of transactions across all payment methods, indicating stronger purchasing activity. These insights help businesses optimize payment infrastructure, improve customer convenience, and develop payment strategies that align with customer preferences.

Business Question 10: How is the distribution of the payment method?

SQL Query:

```
SELECT payment_method,  
       COUNT(invoice_no) AS total_transactions  
FROM customer  
GROUP BY payment_method;
```

SQL Output:

Payment Method	Total Transactions
Cash	420
Credit Card	366
Debit Card	214

This query groups customer transactions by payment method and counts the total number of transactions for each payment type. It helps identify the most preferred payment methods and provides insights into customer payment behavior.

Business Interpretation:

The analysis shows that **Cash** is the most preferred payment method with **420 transactions**, followed by **Credit Card (366 transactions)** and **Debit Card (214 transactions)**. This indicates that customers primarily prefer cash payments, while card payments also account for a significant share of transactions. These insights help businesses optimize payment infrastructure, improve checkout efficiency, and develop payment strategies that align with customer preferences.

Business Question 11: Which are the top 5 shopping malls by revenue? (Extra Query)

SQL Query:

```
SELECT shopping_mall,
       ROUND(SUM(revenue),2) AS total_revenue
FROM customer
GROUP BY shopping_mall
ORDER BY total_revenue DESC
LIMIT 5;
```

SQL Output:

Rank	Shopping Mall	Total Revenue
1	Mall of Istanbul	528,002.21
2	Kanyon	377,836.12
3	Metrocity	336,084.12
4	Metropol AVM	266,128.68
5	Istinye Park	181,847.34

This query groups customer transactions by **shopping mall**, calculates the total revenue generated by each location, and ranks the malls in descending order of revenue. It helps identify the highest-performing shopping malls and supports location-based business performance analysis.

Business Interpretation:

The analysis shows that **Mall of Istanbul** generated the highest revenue (**528,002.21**), followed by **Kanyon**, **Metrocity**, **Metropol AVM**, and **Istinye Park**. These shopping malls represent the top-performing business locations based on revenue generation. The findings help businesses identify high-performing locations, optimize resource allocation, prioritize marketing investments, and evaluate opportunities for expanding successful business operations.

11. Overall SQL Analysis – Key Findings

11.1 Overview:

The SQL analysis explored customer purchasing behavior, sales performance, demographic trends, payment preferences, and shopping mall performance. By answering 11 business-focused questions, the analysis identified key patterns that support data-driven decision-making and formed the foundation for the interactive Power BI dashboards.

11.2 Key Findings:

- Female customers generated higher total revenue and purchased more products than male customers, indicating stronger overall purchasing activity.
- Customers aged 50 and above recorded the highest number of transactions, purchased the most products, and generated the highest revenue, making them the most valuable customer segment.
- Clothing emerged as the highest-performing product category across multiple customer segments, contributing significantly to total sales.
- Purchasing behavior varied across age groups and genders, highlighting opportunities for targeted marketing and customer segmentation.
- Cash was the most preferred payment method, followed by Credit Card and Debit Card, reflecting customer payment preferences.
- Female customers recorded more transactions than male customers across all payment methods, indicating consistent purchasing activity.
- Mall of Istanbul generated the highest revenue among all shopping malls, followed by Kanyon, Metrocity, Metropol AVM, and Istinye Park.
- Revenue distribution across shopping malls highlighted high-performing business locations that can be prioritized for marketing investments and business expansion.
- SQL analysis successfully transformed raw customer transaction data into meaningful business insights that support strategic planning and operational decision-making.

11.3 Business Value:

The SQL analysis provides actionable insights that help businesses:

- Identify high-value customer segments.
 - Understand customer purchasing behavior.
 - Optimize inventory planning based on product demand.
 - Improve customer segmentation and targeted marketing.
 - Enhance payment service strategies.
 - Evaluate shopping mall performance and prioritize high-performing locations.
 - Support data-driven business decisions using reliable analytical insights.
-

12. Power BI Dashboard Analysis

Following the SQL-based analysis, the cleaned dataset was imported into **Power BI** using **Power Query**, where the data model was prepared for visualization. **DAX measures** were created to calculate key business metrics and develop two interactive dashboards that provide interactive and actionable business insights.

The **Executive Overview Dashboard** provides senior stakeholders with a high-level summary of business performance through key performance indicators (KPIs), revenue trends, customer demographics, product performance, and payment insights. The **Business Insights Dashboard** offers a more detailed analytical view, focusing on customer segments, shopping mall performance, product category analysis, age group trends, and other business metrics that support operational and strategic decision-making.

Both dashboards include interactive slicers for **Gender**, **Age Group**, **Category**, **Shopping Mall**, and **Payment Method**, enabling users to dynamically filter the data and explore business insights from multiple perspectives. This interactive design allows users to quickly identify trends, compare performance across different customer segments, and make informed business decisions.

12.1 Dashboard 1 - Executive Overview

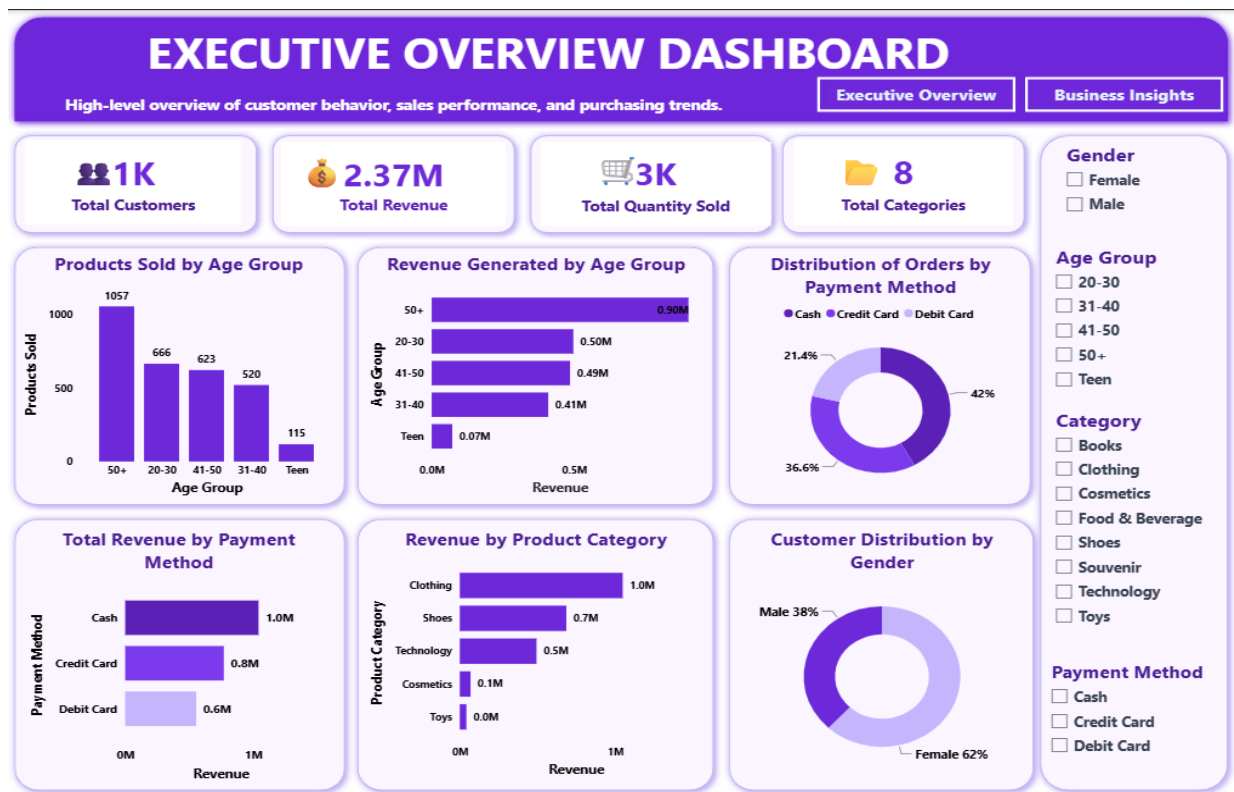


Figure 1: Executive Overview Dashboard

KPI Cards

The dashboard presents four key performance indicators that provide an immediate overview of business performance:

- Total Customers:** Approximately **1,000** customers.

- **Total Revenue: 2.37 Million.**
- **Total Quantity Sold:** Approximately **3,000** units.
- **Total Categories: 8** product categories.

These KPIs provide stakeholders with a quick summary of customer activity, sales performance, and overall business scale before exploring detailed visualizations.

Revenue Analysis

Revenue performance is analyzed by **Age Group** and **Payment Method**.

The **Revenue Generated by Age Group** chart shows that customers aged **50+** generated the highest revenue (**0.90 Million**), followed by the **20–30** and **41–50** age groups. The **Teen** segment contributed the lowest revenue. The **Revenue by Payment Method** chart shows that **Cash** generated the highest revenue, followed by **Credit Card** and **Debit Card**. These insights help identify the customer segments and payment channels that contribute most to overall business revenue.

Customer Demographics

Customer demographics are analyzed using the **Customer Distribution by Gender** donut chart and the **Products Sold by Age Group** column chart.

The dashboard shows that **62%** of customers are female and **38%** are male. Among all age groups, customers aged **50+** purchased the highest number of products (**1,057 units**), followed by the **20–30**, **41–50**, and **31–40** age groups, while the **Teen** segment recorded the lowest purchase volume. These insights help businesses understand customer demographics and identify the most valuable customer segments.

Product Category Analysis

The **Revenue by Product Category** chart highlights the sales performance of each product category. **Clothing** generated the highest revenue (**approximately 1.0 million**), followed by **Shoes (0.7 million)**, **Technology (0.5 million)**, **Cosmetics (0.1 million)**, and **Toys**, which contributed the lowest revenue. These insights help identify the most profitable product categories and support inventory planning, product assortment, and promotional strategies.

Payment Method Analysis

The **Distribution of Orders by Payment Method** donut chart shows that **Cash** accounts for **42%** of all transactions, followed by **Credit Card (36.6%)** and **Debit Card (21.4%)**. Combined with the **Total Revenue by Payment Method** chart, the analysis confirms that Cash is both the most frequently used payment method and the highest revenue-generating payment channel. These insights help businesses optimize payment services and improve the customer checkout experience.

Interactive Segmentation Filters

The dashboard includes interactive slicers for **Gender**, **Age Group**, **Category**, **Shopping Mall**, and **Payment Method**, allowing users to dynamically filter all visuals. This functionality enables stakeholders to compare customer segments, analyze purchasing patterns, and explore business performance from multiple perspectives without leaving the report.

12.2 Dashboard 2 - Business Insights

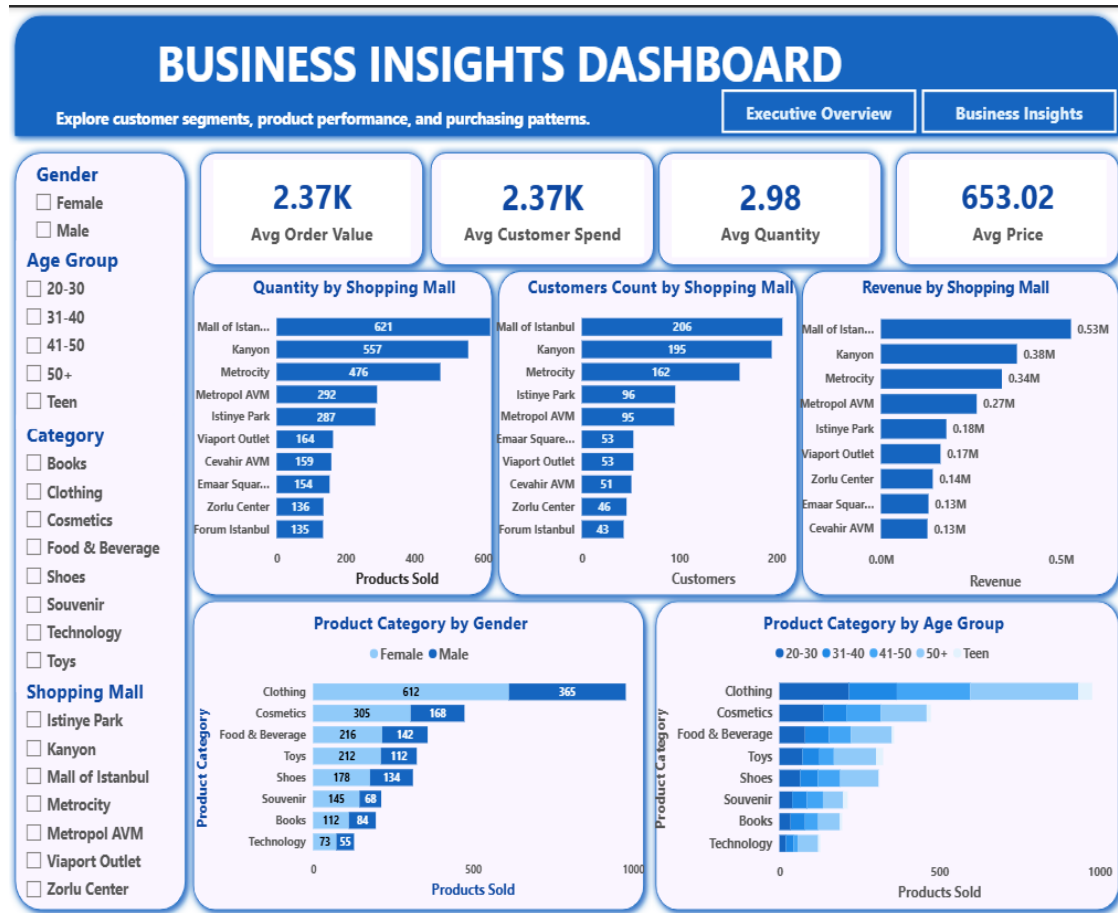


Figure 2: Business Insights Dashboard

Average Order Value, Average Customer Spend, Average Quantity, and Average Price

The dashboard presents four KPI cards that summarize customer purchasing behavior and transaction performance. The average order value and average customer spend are approximately **2.37K**, while the average quantity purchased per transaction is **2.98 units** and the average product price is **653.02**. These KPIs provide a quick overview of customer spending patterns and overall sales performance.

Shopping Mall Performance

The dashboard evaluates shopping mall performance using **Quantity Sold**, **Customer Count**, and **Revenue**. **Mall of Istanbul** ranks first across all three metrics, followed by **Kanyon**, **Metrocity**, **Metropol AVM**, and **Istinye Park**. This consistent performance indicates that these locations are the strongest contributors to overall business growth. These insights support location-based marketing, resource allocation, and business expansion decisions.

Customer Segmentation

Interactive slicers for **Gender**, **Age Group**, **Category**, and **Shopping Mall** enable users to filter dashboard visuals dynamically. This functionality allows stakeholders to analyze purchasing behavior across different customer segments, compare performance, and gain deeper business insights without modifying the underlying report.

Product Category by Gender

The chart compares product purchases between female and male customers across all product categories. **Clothing** is the highest-selling category for both genders, with **612 units** purchased by female customers and **365 units** by male customers. Female customers purchased more products than male customers in every category, indicating stronger overall purchasing activity.

Product Category by Age Group

The stacked bar chart illustrates product purchases across different age groups. **Clothing** is the highest-performing category, with customers aged **50+** and **20–30** contributing the largest purchase volumes. This analysis highlights variations in product preferences across age groups and supports targeted marketing, inventory planning, and customer segmentation strategies.

Dashboard Summary

The Business Insights Dashboard provides a detailed analytical view of customer behavior, shopping mall performance, and product sales across different demographic segments. Through interactive visualizations and dynamic filtering capabilities, the dashboard enables stakeholders to identify business trends, evaluate performance, and support informed decision-making for marketing, inventory management, and strategic business planning.

This creates a smooth transition to the **Overall Business Insights** section and gives your report a polished, professional ending before the final conclusions.

13. KPI Summary

The table below summarizes all Key Performance Indicators (KPIs) used across the Executive Overview and Business Insights dashboards. Each KPI is accompanied by its calculated value and a brief explanation of its business significance, helping stakeholders understand how the metrics support performance evaluation and decision-making.

KPI	Value	Explanation
Total Customers	1K	Represents the total number of unique customers included in the dataset, providing an overview of the customer base analyzed.
Total Revenue	2.37M	Represents the total revenue generated from all customer transactions, measuring the overall sales performance of the business.
Total Quantity Sold	3K	Represents the total number of product units sold across all customer transactions, indicating overall sales volume.
Total Categories	8	Represents the total number of product categories available in the dataset, reflecting the diversity of products offered.
Average Order Value	2.37K	Represents the average revenue generated per customer transaction, helping evaluate the typical value of each purchase.
Average Customer Spend	2.37K	Represents the average amount spent by each customer during the analysis period, providing insight into customer purchasing behavior.
Average Quantity	2.98	Represents the average number of product units purchased per transaction, indicating the typical basket size.
Average Price	653.02	Represents the average selling price of products across all transactions, providing insight into the overall pricing level.

KPI Summary Insight

The KPIs collectively provide a high-level overview of business performance by summarizing customer activity, revenue generation, sales volume, product diversity, and purchasing behavior. Together, these metrics enable stakeholders to monitor business performance, identify trends, and support data-driven decision-making.

14. Business Insights

This section summarizes the key business insights derived from the SQL analysis and the Power BI dashboards. These findings highlight customer behavior, product performance, payment preferences, and shopping mall performance, providing actionable recommendations to support data-driven business decision-making.

14.1 Customer Demographics

Female customers represent **62%** of the customer base and consistently generate higher purchasing activity than male customers across all product categories. Customers aged **50 and above** are the most valuable customer segment, leading in transaction count, product purchases, and revenue generation. These findings suggest that marketing campaigns, product offerings, and customer retention strategies should prioritize this high-value customer segment.

14.2 Product Performance

Clothing is the highest-performing product category across all customer segments, making it the primary driver of business revenue. **Shoes** and **Technology** also contribute significantly to overall sales, while **Books**, **Souvenirs**, and **Toys** generate comparatively lower revenue. These insights support inventory optimization, product assortment planning, and targeted promotional strategies.

14.3 Payment Behavior

Cash is the most preferred payment method, accounting for **42%** of all customer transactions and generating the highest revenue. **Credit Card** is the second most popular payment option, followed by **Debit Card**. Understanding customer payment preferences enables businesses to optimize payment infrastructure, improve the checkout experience, and develop effective payment-related strategies.

14.4 Shopping Mall Performance

Mall of Istanbul is the highest-performing shopping mall based on revenue, customer count, and product sales, followed by **Kanyon**, **Metrocity**, **Metropol AVM**, and **Istinye Park**. The performance gap between these leading malls and the remaining locations highlights opportunities to improve sales, customer engagement, and marketing effectiveness at lower-performing shopping malls.

Overall Business Insight

The combined SQL analysis and Power BI dashboards reveal clear patterns in customer demographics, purchasing behavior, product performance, payment preferences, and shopping mall performance. These insights enable businesses to make informed decisions regarding customer segmentation, inventory management, marketing strategies, and operational planning, ultimately supporting sustainable business growth and improved decision-making.

15. Recommendations

Based on the insights derived from the SQL analysis and Power BI dashboards, the following recommendations are proposed to improve customer engagement, product performance, marketing effectiveness, payment experience, and shopping mall operations. These recommendations support data-driven decision-making and long-term business growth.

15.1 Customer Engagement

Since **female customers** and the **50+ age group** represent the most valuable customer segments, loyalty programs and personalized marketing campaigns should prioritize their preferences, particularly for **Clothing, Shoes, and Technology**. At the same time, targeted campaigns should be introduced to increase engagement among younger customers, especially the **Teen** segment, to encourage long-term customer growth.

15.2 Product Sales

As **Clothing, Shoes, and Technology** generate the majority of revenue, inventory planning and supplier management should prioritize these categories to ensure consistent product availability. Lower-performing categories such as **Books** and **Toys** should be reviewed through promotional campaigns, pricing strategies, and improved product placement to increase customer interest and sales.

15.3 Marketing Strategy

Marketing investments should focus primarily on **female customers** and the **50+ age group**, as they contribute the largest share of transactions and revenue. Category-specific campaigns should highlight **Clothing**, while separate promotional strategies can be developed to improve engagement among male customers and younger age groups.

15.4 Shopping Mall Performance

The operational practices of the highest-performing shopping malls—**Mall of Istanbul, Kanyon, and Metrocity**—should be evaluated and adopted where appropriate across lower-performing locations. Regular monitoring of **revenue, customer count, and quantity sold** will help identify performance gaps and support continuous operational improvement.

15.5 Payment Experience

While **Cash** remains the most preferred payment method, businesses should continue to maintain efficient cash-handling processes while encouraging greater adoption of digital payment methods through loyalty rewards, cashback offers, or promotional discounts. This approach can improve customer convenience while reducing operational costs over time.

Summary:

The recommendations presented in this report are based on comprehensive SQL analysis and interactive Power BI dashboards. Implementing these strategies can help improve customer engagement, optimize product performance, enhance operational efficiency, and support sustainable business growth through informed, data-driven decision-making.

16. Business Conclusion

This analysis of 1,000 customer transactions demonstrates that the business's revenue and engagement are concentrated in identifiable, addressable patterns rather than spread evenly across the customer base. Female customers and shoppers aged 50 and above drive a disproportionate share of both transaction volume and revenue; Clothing, Shoes, and Technology are the categories that matter most to the bottom line; Cash remains the dominant payment method; and a small group of shopping malls - led by Mall of Istanbul - accounts for the strongest performance across every location-level measure.

The combination of SQL-based analysis and interactive Power BI dashboards provides stakeholders with both detailed analytical insights and an intuitive platform for exploring business performance. Together, these tools support data-driven decision-making in customer segmentation, product management, marketing strategy, payment optimization, and shopping mall performance, enabling the business to make informed strategic decisions and drive sustainable growth.

17. Future Scope

Several extensions to this analysis would further strengthen the business's ability to act on the findings presented in this report:

- Incorporate a time-based trend view using the existing `invoice_date` field to analyze revenue and quantity by month or season, enabling the identification of seasonal demand patterns.
- Introduce customer-level repeat-purchase analysis to distinguish one-time buyers from repeat customers, which would refine the interpretation of average customer spend.
- Expand the dataset beyond 1,000 records where possible, to increase statistical confidence in segment-level and mall-level comparisons.
- Add profitability data (cost of goods sold) alongside revenue to allow margin-based analysis rather than revenue-based analysis alone.
- Automate the data refresh pipeline from the source system into MySQL and Power BI to keep the dashboards current without manual re-import.

18. References

SQL Project File: Customer Data Analysis SQL script developed in MySQL Workbench, including database creation, data cleaning, feature engineering, and SQL business analysis queries.

Power BI Dashboard: Executive Overview Dashboard and Business Insights Dashboard developed using Power Query for data transformation and DAX for KPI calculations and interactive visualizations.

Tools & Technologies: MySQL, SQL, Microsoft Power BI, Power Query, Data Analysis Expressions (DAX).

Dataset: Customer Shopping Dataset (used for customer behavior, sales performance, payment analysis, product category analysis, and shopping mall performance).

Finding	Recommendation
Female customers contribute more revenue	Focus loyalty programs on this segment
50+ customers generate highest revenue	Design age-specific campaigns
Clothing leads sales	Maintain inventory and promotions
Cash is most used	Encourage digital payments with incentives
Mall of Istanbul performs best	Replicate successful practices at other malls